

A Source-Controlled Nonlinear Kapitza Model

Exact reduction, imperfect pitchfork unfolding, and reproducible
computation

Discussion draft

July 22, 2026

The physical control parameter is the source strength

Two-layer Poisson problem

$$\begin{aligned}\Omega_I &= [-1, 0], & \Omega_E &= [0, 1], \\ -\lambda u_I'' &= 0, & -\mu u_E'' &= 2\mu\kappa, \\ u_I(-1) &= u_E(1) = 0.\end{aligned}$$

Interpretation

- ▶ λ, μ : material conductivities;
- ▶ $\delta = \lambda/\mu$: conductivity ratio;
- ▶ κ : normalized source strength;
- ▶ source amplitude after division by μ : 2κ .

The bulk fields are explicit

Imposing the exterior boundary conditions gives

$$u(z) = \begin{cases} a(z+1), & z \in \Omega_I, \\ b(z-1) + \kappa(1-z^2), & z \in \Omega_E. \end{cases}$$

Interface traces

$$u_I(0) = a, \quad u_E(0) = \kappa - b.$$

Transmission law

$$\lambda a = \mu b = \beta_{\text{phys}}(\kappa - b - a).$$

Elimination gives one scalar fixed-point problem

Flux continuity gives $b = \delta a$, hence

$$s = [u] = \kappa - (1 + \delta)a.$$

After normalizing the conductance and heat flux by μ ,

$$\hat{q} = \delta a = \hat{\beta}(s)s.$$

Therefore

$$a = F_{\kappa,h}(a) = \frac{\kappa \hat{\beta}_{\kappa,h}(s)}{\delta + (1 + \delta) \hat{\beta}_{\kappa,h}(s)}.$$

The physical interval moves with the control parameter:

$$0 < a < \frac{\kappa}{1 + \delta}, \quad 0 < s < \kappa.$$

The new exponential contact law is exactly solvable

Choose

$$\hat{\beta}_{\kappa,h}(s) = \frac{\delta}{1+\delta} \exp[2(\kappa - 2s) + 2h].$$

Define

$$x = \frac{2(1+\delta)a}{\kappa} - 1 = 1 - \frac{2s}{\kappa}.$$

Since $\kappa - 2s = \kappa x$, the complete steady equation becomes

$$x = \tanh(\kappa x + h).$$

The source parameter now controls both the moving physical scale and the normalized sigmoid slope.

The critical source strength is one

For $h = 0$, the equation is $x = \tanh(\kappa x)$.

Root count

$0 < \kappa \leq 1$: 1 state,
 $\kappa > 1$: 3 states.

$$\boxed{\kappa_c = 1.}$$

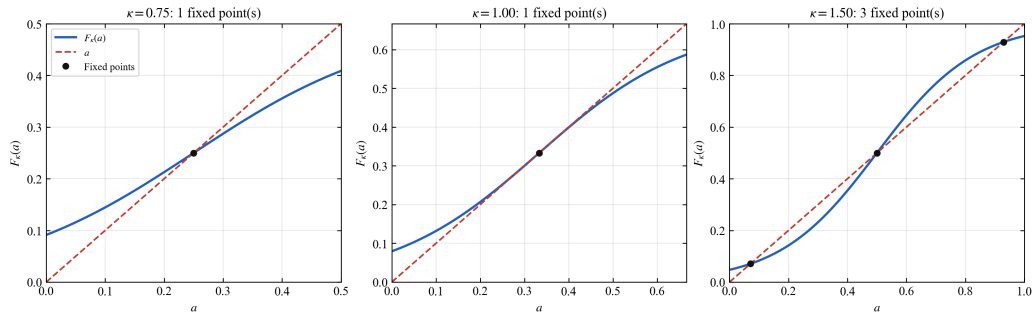
Exact branches

$$a_c(\kappa) = \frac{\kappa}{2(1 + \delta)},$$

$$\kappa(x) = \frac{\operatorname{artanh} x}{x}, \quad a(x) = \frac{\kappa(x)(1 + x)}{2(1 + \delta)}.$$

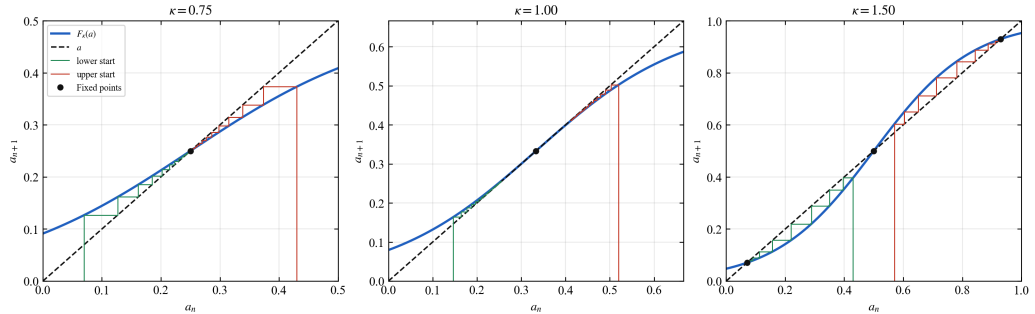
Fixed-point intersections show the transition

Fixed-point geometry as the source strength crosses $\kappa_c = 1$



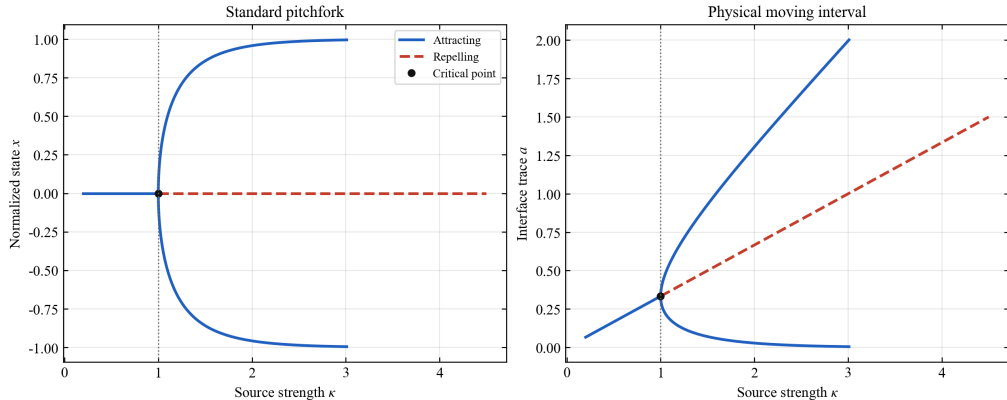
Cobwebs reveal critical slowing and branch selection

Cobweb iterations reveal stability and branch selection



The physical pitchfork has a moving centerline

Supercritical pitchfork generated by $x = \tanh(\kappa x)$



Stability means attraction of the Picard iteration

At a fixed point,

$$m_{\text{FP}} = \kappa \operatorname{sech}^2(\kappa x + h) = \kappa(1 - x^2).$$

Line convention

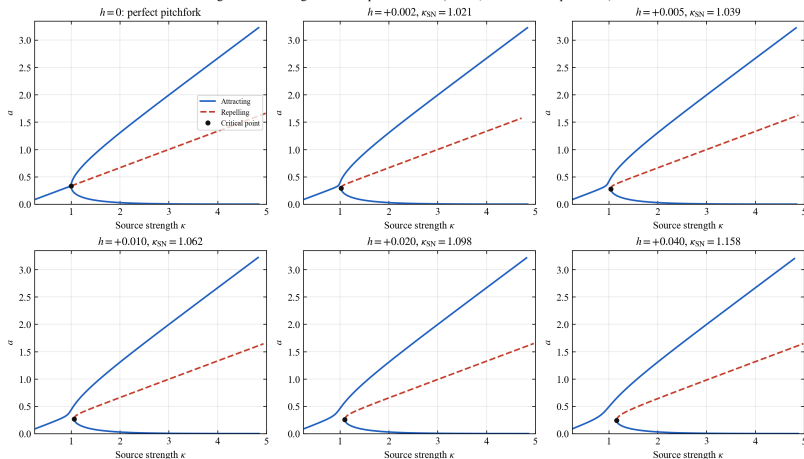
- ▶ solid: $|m_{\text{FP}}| < 1$, attracting;
- ▶ dashed: $|m_{\text{FP}}| > 1$, repelling.

Scope

This is stability of the fixed-point algorithm. Physical transient stability requires a separate time-dependent interface model.

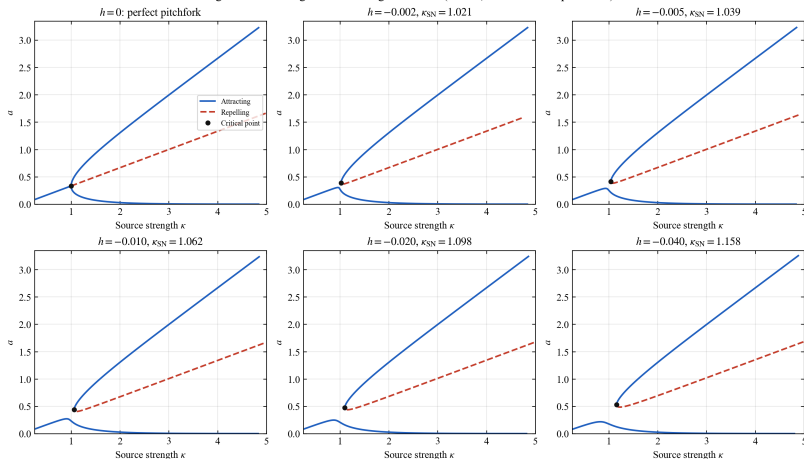
Small positive biases unfold the pitchfork

Progressive unfolding at six small positive biases ($\delta=0.5$; κ is the control parameter)



Negative biases produce the reflected unfolding

Progressive unfolding at six small negative biases ($\delta=0.5$; κ is the control parameter)



The saddle-node trajectory is analytic

The fold equations are

$$\kappa_{\text{SN}} = \frac{1}{1 - x_{\text{SN}}^2}, \quad h = \operatorname{artanh} x_{\text{SN}} - \frac{x_{\text{SN}}}{1 - x_{\text{SN}}^2}.$$

Near $(\kappa, h) = (1, 0)$,

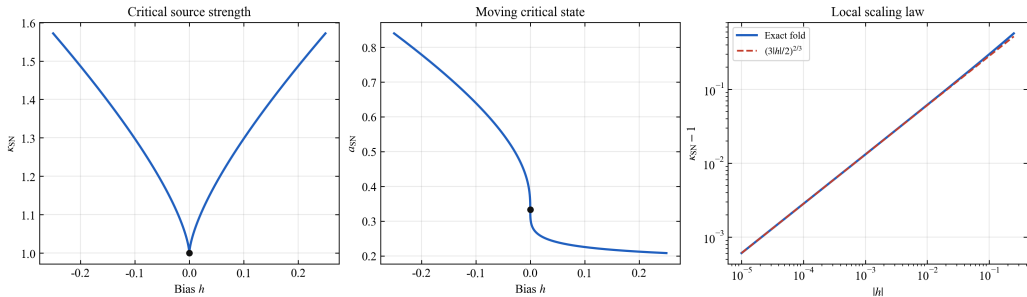
$$h = -\frac{2}{3}x_{\text{SN}}^3 + O(x_{\text{SN}}^5), \quad \kappa_{\text{SN}} - 1 = x_{\text{SN}}^2 + O(x_{\text{SN}}^4),$$

hence

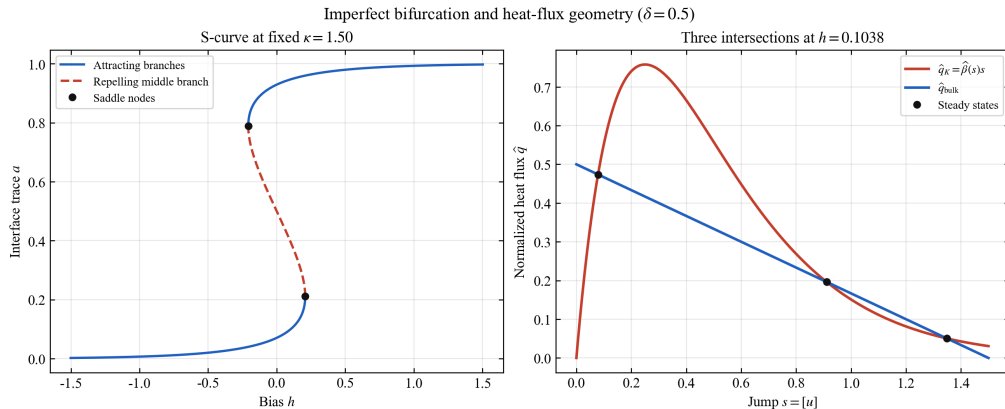
$$\kappa_{\text{SN}} - 1 \sim \left(\frac{3|h|}{2} \right)^{2/3}.$$

The critical point follows a cusp trajectory

Saddle-node trajectory created by imperfect pitchfork unfolding

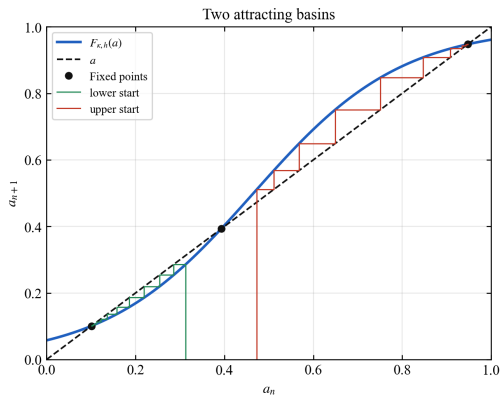
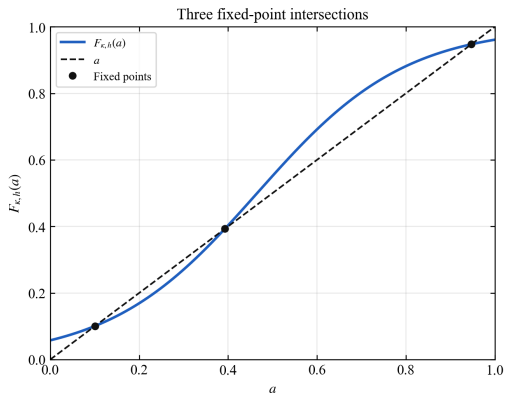


Fixed κ : S-curve and equivalent flux roots

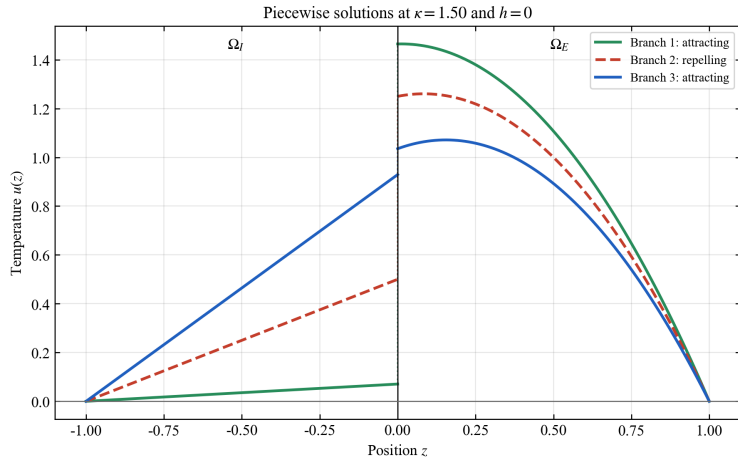


A fixed-bias slice has two attracting basins

Fixed- h slice: $\delta = 0.5, \kappa = 1.50, h = 0.1038$



Every scalar root reconstructs a complete temperature field



The implementation is reproducible in two languages

Python

code/python/
kapitza_kappa_model.py

- ▶ one self-contained script;
- ▶ bracketed root isolation;
- ▶ PNG, vector PDF, and CSV outputs.

MATLAB

code/matlab/
kapitza_kappa_model.m

- ▶ matching equations and figures;
- ▶ solid/dashed stability convention;
- ▶ English comments throughout.

PDE reduction from the cited paper; exponential conductance is the new constitutive construction.

Thank you

Questions and discussion